

Assignment 2

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Task 1

Task 1.a

$$d_e = 24, \quad d_m = 3.14606742$$

Task 1.b

Task 1.b.i

If every observation is multiplied by 10, average will be also multiplied by 10. However, the average of v_1 is 0, so nothing changes in $\bar{v}$:

$$\bar{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

v_0 is a single observation, so its first component will be multiplied by 10, nothing else changes:

$$\bar{v}_0 = \begin{bmatrix} 20 \\ 5 \\ 3 \end{bmatrix}$$

As for S, it will look like this:

$$S = \begin{bmatrix} 500 & 10 & 20 \\ 10 & 6 & 1 \\ 20 & 1 & 7 \end{bmatrix}.$$

S_{11} element is a variance of v_1 . v_1 is increased in 10 times, so the variance is increased in $10^2 = 100$ times. $S_{12} = S_{21}$ and $S_{13} = S_{31}$ are the covariances of v_1 with v_2 and v_3 correspondingly. v_2 and v_3 do not change, so the increase of v_1 in 10 times means that covariances increase in 10 times.

Other elements of S does not include v_1 , so they are unchanged.

Task 1.b.ii

$$d_e = 420, \quad d_m = 3.14606742$$

d_e increased, which is reasonable, because it is simply a sum of squared distances for each component.

d_m remains unchanged, because it is additionally normalized using the variances and covariances of the components.

Task 2

Task 2.a

First, we can represent X as the vector of column-vectors:

$$X = [1, \ln(X_2), X_3]$$

Similarly with X^* :

$$X^* = [1, \ln(a \cdot X_2), b \cdot X_3]$$

If X and X^* are $n \times 3$, A should be 3×3 . Moreover, to get i^{th} element of the first column of X^* we need to multiply the i^{th} row of X with the first column of A . Therefore, in the first column of A the first element should be 1, and other elements should be zeros.

Similarly, the third element of the third column should be b , and other elements in this column must be zeros.

As for the second column, it should transform from $\ln(X_2)$ to $\ln(a \cdot X_2) = a + \ln(X_2)$, so in the first row it will be a (adding constant), in the second it will be 1, and third is 0.

Therefore:

$$A = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & b \end{bmatrix}.$$

Task 2.b

$$\begin{aligned} \hat{\beta}^* &= (X^{*\top} X^*)^{-1} X^{*\top} y \\ &= ((XA)^\top XA)^{-1} (XA)^\top y \\ &= (A^\top X^\top X A)^{-1} A^\top X^\top y \\ &= A^{-1} (X^\top X)^{-1} (A^\top)^{-1} A^\top X^\top y \\ &= A^{-1} (X^\top X)^{-1} X^\top y \\ &= A^{-1} \hat{\beta}. \end{aligned}$$

So, the relation of $\hat{\beta}^*$ and $\hat{\beta}$ equals to A^{-1} .

$$A^{-1} = \begin{bmatrix} 1 & -a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{b} \end{bmatrix}$$